

Milestone 3

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Agentic AI

Overview

This milestone implements a multi-agent workflow for my weight-loss planning assistant. The goal of the system is to help users create a personalized weight-loss plan based on their inputs and preferences. This system update relies on a system divided into multiple agents, which will improve the structure, reliability, and clarity of the outputs. The updated workflow follows a fixed pipeline that has a planner agent, a calculator agent, and an evaluator agent. Additionally There is a revision loop that will improve outputs if it doesn't meet the standards.

Agent Roles and Responsibilities

1. Planner agent

The planner agent is responsible for organizing the users' inputs and generating a personalized plan. This agent transforms raw user data into a usable format. The planner agent takes in user inputs such as current weight, goal weight, height, meal preferences, exercise preferences, and desired timeline. Using user inputs, the system produces a user profile along with a draft weekly meal plan and a workout plan. The agent also identifies the need for a caloric deficit, which will be used hand in hand with the calculator agent.

Input:

- Current weight
- Goal weight
- Height
- Meal preferences
- Exercise preferences
- Desired timeline
- Fitness goal

Output:

- User profile summary
- Weekly meal plan structure
- Weekly workout plan structure
- Assumptions for calculation

2. Calculator Agent

The calculator agent is responsible for performing numerical estimations related to the user's weight-loss goals. It uses the structured output from the

planner agent to generate weight-loss projections. The calculator agent estimates the weekly rate of weight loss and provides a reasonable timeline for achieving the user's weight-loss goals. This also tells the user whether the desired timeline is realistic or too aggressive, and whether it is not possible within that timeline.

Input:

- Planner Agent output (user profile and assumptions)

Output:

- Estimated weekly weight loss rate
- Estimated timeline to reach goal weight
- Realism assessment (reasonable or aggressive)

3. Evaluator Agent

The evaluator Agent is responsible for verifying the completeness, safety, and quality of the output generated by previous agents. It ensures that the final result is useful and in an appropriate format for the user. This agent ensures the plan is personalized using all user inputs, whether the timeline and recommendations are realistic, and the finalized response is clear. This evaluator also checks that no harmful advice is given. If the output meets the required standards, the evaluator returns approval; if not, it returns specific feedback indicating what needs improvement.

Input:

- Planner Agent output
- Calculator Agent output
- Original user inputs

Output:

- Approved
- Unapproved and revised feedback

Communication Between Agents

The three agents communicated via a fixed pipeline, with each agent passing structured information to the next. This ensures that each step builds on the others without losing important context or information.

The planner agent retains raw user input and transforms it into a structured plan. This structured plan is then used by the calculator agent to calculate the estimation. Lastly, the evaluator agent receives both the planner agent's structured plan and the calculation agents' estimates, and compares them against the initial user's input.

The evaluator agent also approves or disapproves the output and sends the feedback back to the planner agent. The planner agent will then update the plan, and the revised output will be sent to each agent again. This loop ensured consistency and clarity, since each agent has a clearly defined role.

Failure Handling and Quality Control

The system has a built-in mechanism for handling low-quality or unrealistic outputs. This is where the evaluator agent kicks in and revises the output.

The evaluator agent checks for issues that may include:

- unrealistic timelines (e.g., attempting to lose too much weight too quickly)
- lack of personalization
- incomplete or unclear recommendations
- unsafe or extreme guidance

The evaluator reviews the output and, if any of these issues are detected, the evaluator agent revises it and provides specific feedback indicating exactly what needs to be changed. The system then loops this revised output back to the planner agent, which then updates the plan.

The system has a stopping condition of a maximum of two revision attempts, thereby preventing an infinite loop. If the output is still not approved after the second attempt, the system will return the best available version along with notes.